

Ph.D. Student Status Update and Self Evaluation

School of Computing, UGA, Spring 2026

Last Name, First Name: BISHI, Rahman KOREDE

Last four digits of UGA ID: 3422

Semester and Year when you joined the Ph.D. program[±]: Fall, 2022

Date when this form is being submitted: 04/15/2026

Name of the Major Professor if Chosen (write "NC" if not yet chosen): Dr John Miller

Names of the PhD Advisory Committee if the Committee has been Formed (write "NC" if the committee has not yet been formed):

Committee Member: Dr. John A. Miller (Major Professor)

Committee Member: Dr. Qianwen Li

Committee Member: Dr. Ingrid Maria Hybinette

MILESTONES:

(For each milestone listed below, if you have met the milestone provide the semester and year when it was met. If you have not yet met the milestone write "NM", and if you have a reasonable estimate as to when (semester & year) the milestone would be met please indicate in parenthesis).

1. **Formation of PhD advisory committee (completion of the G130 Form):** Fall 2025
(Under normal circumstances, the student is expected to meet this milestone by the 3rd semester not counting summer semesters)

2. **Core competency:** Fall 2025
(Under normal circumstances, the student is expected to meet this milestone by the end of the 3rd semester not counting summer semesters)

3. **Preliminary Doctoral Program of Study Form:** Fall 2025
(Under normal circumstances, the student is expected to meet this milestone by the 3rd semester not counting summer semesters)

4. **Paper Submission before Comprehensive Exam:** Fall 2025
(Under normal circumstances, the student is expected to meet this milestone during the 5th semester not counting summer semesters)

a. **Title: BEYOND CORRIDOR AVERAGES: LANE-LEVEL VALIDATION OF MICROSCOPIC FREEWAY SIMULATION WITH EMPIRICALLY PARAMETERIZED ARRIVALS**

b. **Conference/Journal: Annual Modeling and Simulation Conference (ANNSIM 2026)**

5. **Final Program of Study Form (G138):** Fall 2025
(Under normal circumstances, the student is expected to meet this milestone during the 5th semester not counting summer semesters)

6. **Comprehensive Exam:**
(Under normal circumstances, the student is expected to meet the below two milestones by the end of the 6th semester)

a. **Written Exam:** In progress -- Spring, 2026: Current

b. **Oral Exam:** Waiting Written Result NM (Summer, 2026)

7. **Application for Admission to Candidacy:** NM (Summer, 2026)
(Under normal circumstances, the student is expected to meet this milestone by the end of the 6th semester not counting summer semesters)

8. **Paper Accepted Before Prospectus:** Fall 2025
(Under normal circumstances, the student is expected to meet this milestone during the 7th semester not counting summer semesters)

a. **Title: BEYOND CORRIDOR AVERAGES: LANE-LEVEL VALIDATION OF MICROSCOPIC FREEWAY SIMULATION WITH EMPIRICALLY PARAMETERIZED ARRIVALS**

b. **Conference/Journal:**
ANNSIM2026

9. **Prospectus:** NM (Fall, 2026)
(Under normal circumstances, the student is expected to meet this milestone by the end of the 8th semester not counting summer semesters)

10. **Application for Graduation Form:** NM (Spring, 2028)

(Under normal circumstances, the student is expected to meet this milestone by the beginning of the 10th semester not counting summer semesters)

11. Defense: _____ **NM (Spring, 2028)** _____

(Under normal circumstances, the student is expected to meet this milestone by the end of the 10th semester not counting summer semesters)

RESEARCH PROGRESS:

Research Progress Summary:

My dissertation investigates empirically validated microscopic traffic simulation for infrastructure resilience analysis, organized around three studies.

1. **Study 1:** (ANNSIM 2026, accepted) establishes which modeling decisions materially affect simulation accuracy: we show that arrival-process choice has far greater impact than numerical integrator choice -- the shifted Erlang-2 distribution reduced flow prediction error by ~28% compared to Poisson, while integrator choice produced <1% variation.
2. **Study 2:** (in preparation) applies the validated simulation to wildfire evacuation resilience, reconstructing the 2025 Palisades Fire evacuation on I-210 and evaluating contraflow strategies through counterfactual scenarios calibrated against PeMS fire-day sensor data.
3. **Study 3:** (proposed) envisions a unified agentic architecture where LLM-driven agents autonomously design, execute, and refine simulation experiments over a validated digital twin, enabling scenario exploration at a scale infeasible through manual experimentation.

The simulation engine is built within the ScalaTion 2.0 framework and includes three car-following models (IDM, Gipps, Krauss), a MOBIL lane-change model, on-ramp merge behavior, variable-lane doubly-linked-list vehicle tracking, and a Java2D rendering engine with OpenStreetMap background maps. Calibration experiments run on the GACRC Sapelo2 cluster using SPSSA and Nelder-Mead optimizers. I am currently preparing for my written candidacy examination.

List of Publications:

1. Bishi, K.R., Bowman, J., Miller, J.A. (2026). "Beyond Corridor Averages: Lane-Level Validation of Microscopic Freeway Simulation with Data-Driven Arrivals." Annual Modeling and Simulation Conference (ANNSIM 2026). [Accepted]
2. Bishi, K.R., Miller, J.A. (2026). "Evaluating Evacuation Resilience Under Wildfire Disruption: A PeMS-Calibrated Microscopic Simulation of I-210 During the 2025 Palisades Fire." Winter Simulation Conference (WSC 2026). [Manuscript in Preparation]

COURSE WORK PROGRESS:

Number of credit hours completed by end of current semester: 18

Number of credit hours completed by end of current semester that count towards program of study:
0

Coursework completed or to be completed by end of this semester

No. of hours completed from *preliminary focus** by end of this semester: _____

No of hours completed from the *primary focus** area by the end of this semester: _____

No. of hours of 8000-level CSCI coursework: 15

No. of hours of graduate-level CSCI coursework: 0

No. of hours of doctoral minor coursework (if doing a minor) : 3

No. of hours of CSCI 9300 Doctoral Dissertation: _____

* For definitions of the *preliminary focus* and *primary focus* areas please refer to the web page
<http://www.cs.uga.edu/doctor-philosophy-computer-science>

ACTIVITIES DURING CURRENT YEAR (Summer 2025, Fall 2025 and Spring 2026)

Research Activities During this Academic Year Including Publications:

1. Designed and built a microscopic traffic simulation of the US-101 freeway corridor (Willow Road to Marsh Road, 2 miles) within the ScalaTion 2.0 discrete-event simulation framework
2. Implemented three car-following dynamics -- Intelligent Driver Model (IDM), Gipps safe-distance model, and Krauss stochastic model -- with configurable numerical integrators (Euler through Dormand-Prince RK45)
3. Built lane-level validation infrastructure comparing simulated per-lane flow and speed against PeMS sensor data at five detector stations
4. Developed a shifted Erlang-2 arrival process that enforces realistic minimum headways, reducing flow prediction error by 28% versus standard Poisson arrivals
5. Implemented MOBIL lane-change model (Treiber & Kesting, 2007) within the simulation engine, including safety criterion, incentive criterion, and anti-oscillation cooldown
6. Built variable-lane doubly-linked-list architecture supporting segments with different lane counts (e.g., 4-lane mainline with 2-lane on-ramps)
7. Ran calibration experiments on the GACRC Sapelo2 HPC cluster using SPSA optimizer with SLURM array jobs
8. Built a second simulation model for the I-210 / SR-134 corridor (Eaton Fire evacuation study, dual-corridor with on-ramps and off-ramps)
9. Developed a rendering engine using Java2D: road polygons, lane markings, rotated velocity-colored vehicles, OpenStreetMap background map overlay with auto-download, HUD overlay
10. Paper accepted at ANNSIM 2026: "Beyond Corridor Averages: Lane-Level Validation of Microscopic Freeway Simulation with Data-Driven Arrivals"

Course-related Activities During this Academic Year:

- CSCI 9000

- CVLE 8120

Degree-related Plans Milestones Achieved During This Academic Year :

1. Core Competency certified -- July 20, 2025 (Core GPA 3.566, approved by full committee)
2. PhD Advisory Committee formed -- Summer, 2025 (Dr. Miller, Dr. Li, Dr. Hybinette)
3. First paper submitted and accepted -- ANNSIM 2026
4. Written candidacy examination -- in progress (Spring, 2026)

NOTABLE ACHEIVEMENTS/RECOGNITIONS YOU MAY HAVE HAD DURING THIS YEAR
(Best paper awards, TA awards, prizes in hackathons, competitions, etc.)

None

DEVIATIONS FROM PLAN THAT YOU PROVIDED LAST YEAR:

(Summarize any research, coursework, and degree-related milestones that you had planned for the current year but you could not meet them. Please provide reasons. Not needed if you are a 1st year PhD student and haven't submitted the self-evaluation form last year.)

1. Only deviation is Oral Yet to be administered.
2. Awaiting written exam collation

PLANS FOR THE NEXT YEAR:**Research-related Plans:**

I plan to complete my written and oral candidacy examinations by Summer 2026. My next research target is a paper on wildfire evacuation resilience, using the I-210 corridor simulation I have already built to evaluate contraflow strategies during the 2025 Palisades Fire. I will calibrate the model against PeMS fire-day sensor data and run counterfactual scenarios comparing baseline, partial contraflow, and full contraflow configurations. I also plan to continue improving the simulation engine -- gap acceptance at ramp mergers, density-based lane assignment, and further calibration work on Sapelo2. I expect to submit one more paper by end of 2026.

Course Work-Related Plans

I have completed all my required course work.

Degree-related Milestones:

(List degree-related milestones that will be achieved next year)

1. Complete written candidacy exam -- Spring 2026
2. Complete oral candidacy exam -- Summer 2026
3. Admission to candidacy -- Summer 2026
4. Submit second paper -- Fall 2026
5. Prospectus -- Fall 2026 or Spring 2027

FUNDING HISTORY:

In the table below, please enter your funding history as a PhD student. Please follow the guidelines below.

- You should provide this information for every semester you were enrolled as a PhD student. If you changed your degree objective from MS to PhD, provide the information for only the semesters you are enrolled as a PhD student (i.e., do not include semesters you were enrolled as an MS student)
- In Column-1, please enter the semester and year in chronological order (i.e., starting from the first semester you enrolled as a PhD student). Also include summer semesters.
- In column-2 enter the funding type (TA, RA, Self-funded, Internship etc.). If you had multiple funds (e.g., TA and RA), enter both.

- In column-3 enter the unit providing the funds. You may omit the company name if the funding type is internship.
- If you were funded as a RA in a particular semester, in column-4, enter the name(s) of the faculty members that provided the RA.
- Add more rows as needed.

Semester and Year	Funding Type (TA, RA, Self-funded, Internship)	Unit Providing Funding (e.g., SoC, Veterinary College, etc.)	If funded as a RA by faculty member(s) in SoC, please enter their names.
Fall 2022	TA	SoC	
Spring 2023	TA	Soc	
Summer 2023	TA	Soc	
Summer 2023	RA	EETI	Dr Nathaniel Hunsu
Spring 2024	TA	Soc	
Summer 2024	TA	Soc	
Fall 2024	TA	Soc	
Spring 2025	TA	Soc	
Summer 2025	TA	Soc	
Fall 2025	Self-Funded	OGE Partial	
Spring 2026	TA	Soc	

SIGNATURES:

Signature of Student: _____

Date: _____

Signature of Major Professor: _____
(if selected)

Date: _____

For Department Use Only

(Do Not Write)

Reviewed by: _____

Review Date: _____

Review Notes: